

COURSE TITLE : PROCESS CONTROL AND INSTRUMENTATION
COURSE CODE : 6124
COURSE CATEGORY : E
PERIODS PER WEEK : 4
SEMESTER : 6
PERIODS PER SEMESTER : 60
CREDITS : 4

TIME SCHEDULE

MODULE	TOPICS	PERIODS
1	Introduction, Fundamentals of Instrumentation in Pulp, Paper and Wood Industries	14
	Test I	1
2	Process Control variables in Pulp, Paper and Wood Industries Pressure and vacuum Temperature Flow Liquid level Length and speed Density and specific gravity	14
	Test II	1
3	Process control variables Consistency Freeness Basis weight Moisture and humidity Thickness or Caliper PH Optical measurement	14
	Test III	1
4	Process controls in pulp, paper and wood Industries Modern Controls System	14
	Test IV	1
Total		60

Course Outcome :

Module	G.O	Student will be able to:
1	1	Understand the basic concepts of Instrumentation in Pulp, Paper and Wood Industries
2	1	Understand the different process variables in pulp, paper and wood Industries.
3	1	Understand the different process control variables in pulp, paper and wood industries.

4	1	Comprehend the different process control systems
	2	Understand the basic concepts of modern control systems

Specific outcome

MODULE I

1.1.0 Understand the basic concepts of Instrumentation in Pulp, Paper and Wood Industries

- 1.1.1 State measurement and its aim.
- 1.1.2 State applications of instruments
- 1.1.3 Explain various process variables and its importance in wood, pulp and paper Industries
- 1.1.4 Describe the functional elements of measurement system.
- 1.1.5 Describe the performance characteristics of an instrument
- 1.1.6 Explain the basic methods of measurement such as Force and motion balance with figures.
- 1.1.7 State the basic principles used in transducers and transmitters.
- 1.1.8 Describe indicators, recorders and controllers

MODULE II

2.1.0 Understand the different process variables in pulp, paper and wood Industries.

- 2.1.1 Define absolute pressure, atmospheric pressure, gauge pressure and vacuum
- 2.1.2 Describe the working of Bourdon tube pressure gauge and capacitance type pressure gauge with diagrams.
- 2.1.3 Explain the non-electric method of measurement of temperature such as Mercury in Steel and by metallic.
- 2.1.4 Describe the electronic method of temperature measurement such as Thermo couples, Resistance
- 2.1.5 Explain the radiation method of temperature measurement such as pyrometer with figure.
- 2.1.6 State the working principle of venturimeter and Orifice plate rotameter with Diagrams.
- 2.1.7 Explain the open tank AND CLOSED TANK level measurement of liquid with diagram.
- 2.1.8 Explain the working of centrifugal force tachometer.
- 2.1.9 State the method of measuring Electronic speed.
- 2.1.10 List the method of off-line and on-line measurement of Density and specific gravity.

MODULE III

3.1.0 Understand the different process control variables in pulp, paper and wood industries.

- 3.1.1 Define consistency, Explain the different types of on line measurement of consistency such as blade type and rotary type
- 3.1.2 Define freeness , Explain the on line freeness measurement in drainage rate method with diagram

- 3.1.3 Define basis weight or grammage, Describe with sketches the radioactive method of basis weight measurement.
- 3.1.4 Describe with figures the continuous on line measurement of basis weight
- 3.1.5 Explain the different types of moisture measurement such as infrared type and capacitance type
- 3.1.6 Describe the different humidity measurement such as Dew point method wet bulb method and electrical type in wood seasoning kilns
- 3.1.7 Define Hygrometry
- 3.1.8 Describe the method of on line measurement of thickness or caliper of paper by electromagnetic thickness gauge capacitance type
- 3.1.9 Define ph , Compare the measurement of ph by calorimetric and potentiometric method and Describe the control of ph
- 3.1.10 Describe gloss opacity colour formation and Turbidity

MODULE IV

4.1.0 Comprehend the different process control systems

- 4.1.2 Describe process and their control. Controllers and their operation
- 4.1.3 Describe manual and automatic control without mathematical derivation
- 4.1.4 Explain the different types of controls such as proportional, proportional derivative, proportional integral and proportional integral and derivative

4.2.0 Understand the basic concepts of modern control systems

- 4.2.1 State the supervisory and direct control in modern computer system.
- 4.2.2 Compare analogue and digital computer.
- 4.2.3 Explain the basic processing unit, hardware and software of computer.
- 4.2.4 Describe the different types of computers such as analogue computers, digital computers and process computers.
- 4.2.5 Explain the process control by digital computers.

CONTENT DETAILS

MODULE I

Introduction and fundamentals of instrumentation in wood, pulp and paper industries various process variables – measurement and its aim, typical application of instruments, functional elements of measurement systems performance characteristics of instruments- basis methods of measurements – basic principles of transmitters and transducers

MODULE II

Process control variables in pulp paper and wood industries

1. Pressure and vacuum – absolute pressure, atmospheric pressure, gauge pressure, vacuum gauges bourdon tube pressure gauge capacitance type pressure gauge .
2. Temperature – non- electric method – mercury in steel and bi – metallic , Electric method – Thermo – couples, Resistance bulb or RTD

3. Flow – venturitube – orifice plate rotameter
4. Liquid level – open tank level measurement – glass tube float type or dip tube
 - Closed tank level measurement glass tube, capacitance type
5. Length and speed – centrifugal force tachometer electronic speed measurement
6. Density and specific gravity – offline and online measurement

MODULE III

1. Consistency definition – online measurement – blade type rotary type
2. Freeness – definition – online freeness measurement drainage rate method
3. Basis weight – definition – continuous on line measurement of basis weight radioactive method
4. Moisture and humidity definition moisture measurement – infrared type and capacitance type.
Humidity measurement in wood – seasoning kilns – dew point method – wet bulb method – hygrometry – electrical type
5. Thickness or caliper – introduction and online measurement by electromagnetic thickness gauge capacitance type
6. Ph definition measurement of ph by calorimetric method, potentiometric method, control of ph
7. Optical measurement – gloss opacity – colour formation and Turbidity

MODULE IV

Process control in pulp, paper and wood industries

Introduction – manual and automatic control without mathematical derivation

Types of control – proportional, proportional and integral, proportional derivative, proportional integral and derivative Modern control systems

Introduction – supervisory and direct control – types of computers – analogue computer, digital computer basic processing unit and process computers – process and their control, controls and operation.

REFERENCE

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| 1. Measurement and control in paper making | – Robert J, Mc Grill |
| 2. Hand book of pulp and paper technology | - G.A. Smook, CBS |
| 3. Pulp and paper science Vol. I, II, III | – Ronald G Macdonald |
| 4. Instrumentation measurement and analysis | – B.C. Nakra & K. K. Choudhary |
| 5. Process instrumentation | – C.D. Johnson |
| 6. Instrumentation and Process Control | Janardan Prasad, I.K. International. New Delhi |
| 7. Lecture notes | IPIR, Bangalore |
| 8. Paper making systems and their control Vol. I & II | – Francis Bolan. |